Syllabus 9758 Statistics

Example Questions

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1 Permutations and Combinations

1.1 A treasure hunt game is to be played in a 3-storey building. There are 12 specific locations where 5 treasures are to be hidden. Among the 5 treasures, 4 of them are identical. The locations are distributed across the storeys as follows:

- 3 locations on the 1st storey,
- 3 locations on the 2nd storey, and
- 6 locations on the 3rd storey.

Each location can hold at most one treasure.

(a) Find the number of ways to distribute the 5 treasures among the 12 locations for each of the following separate cases:

(i) The 5 treasures can be hidden in any of the 12 locations. [1]

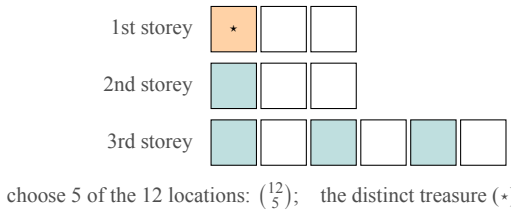
(ii) Exactly 2 treasures must be hidden on the 3rd storey. [2]

(iii) There must be at least one treasure on each of the 1st and 2nd storeys, and at least 2 treasures on the 3rd storey. [3]

Five teams of 2 players each take part in the game. The 10 players and 2 game masters are to be seated around a round table for a briefing.

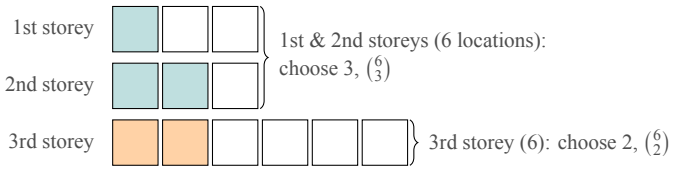
(b) Find the number of different seating arrangements such that players from the same team must be seated next to each other but the two game masters must not be seated next to each other. [3]

[TJC 2025 Prelim P2 Q6]

Solution		Comments
(a)(i) **[1]**	Required number of ways $= \binom{12}{5}\binom{5}{1} = 3960$  choose 5 of the 12 locations: $\binom{12}{5}$; the distinct treasure ($\star$) takes 1 of these 5: $\binom{5}{1}$	

(a)(ii) [2]

Required number of ways $= \binom{6}{2}\binom{6}{3}\binom{5}{1} = 1500$



then the distinct treasure goes to 1 of the 5 chosen locations: $\binom{5}{1}$

Alternative method:

Case 1: 2 not identical at 3rd storey $\binom{6}{2} \times 2! \binom{6}{3} = 600$

Case 2: 2 identical at 3rd storey $\binom{6}{2}\binom{6}{3}\frac{3!}{2!} = 900$

(a)(iii) [3]

Case 1: 2 in 1st storey, 1 in 2nd storey, 2 in 3rd storey

Number of ways $= \binom{3}{2}\binom{3}{1}\binom{6}{2}\binom{5}{1} = 675$

Case 2: 1 in 1st storey, 2 in 2nd storey, 2 in 3rd storey

Number of ways $= \binom{3}{1}\binom{3}{2}\binom{6}{2}\binom{5}{1} = 675$

Case 3: 1 in 1st storey, 1 in 2nd storey, 3 in 3rd storey

Number of ways $= \binom{3}{1}\binom{3}{1}\binom{6}{3}\binom{5}{1} = 900$

Total number of ways $= 2250$

Case	1st (3)	2nd (3)	3rd (6)	ways
1	2	1	2	675
2	1	2	2	675
3	1	1	3	900

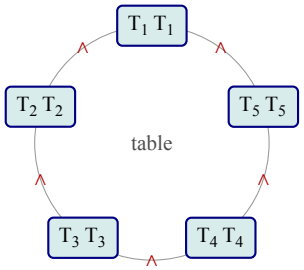
each row: $\binom{3}{\cdot}\binom{3}{\cdot}\binom{6}{\cdot} \times \binom{5}{1}$ (distinct treasure); 5 treasures in total

(b) [3]

Number of ways to arrange the 10 participants $= (5-1)!(2!)^5 = 768$

Number of ways to slot in the 2 game masters $= \binom{5}{2} 2! = 20$

Required number $= 768 \times 20 = 15360$



$(5-1)!$ ways round the table,
$(2!)^5$ within the 5 teams;
game masters in 2 of the 5 gaps ∧:
$\binom{5}{2}$ 2! (never next to each other)

1.2 **(a)** In a survey conducted by a library, a group of 100 members were asked the number of days in a month they visit the library. The results are shown below:

	1 or less	2 to 4	5 or more
Male	15	25	22
Female	20	$18-n$	n

A member is selected at random from the group. The events are defined as follows:

A: The member visits the library on 4 days or less in a month.

B: The member visits the library on 2 days or more in a month.

F: The member is a female.

Find the following probabilities in terms of n.

(i) $\mathrm{P}(A \cup B')$ [1]

(ii) $\mathrm{P}(F \mid A')$ [2]

(iii) Given that $\mathrm{P}(A \cap B) = \frac{3}{10}$, find the value of n. Hence determine if A and B are independent, justifying your answer. [3]

(b) A librarian is organising a large community reading event. To manage registration, he needs to create 4-letter access codes for the participants using the letters from the word BOOKKEEPER.

(i) How many access codes can be formed if all 4 letters are distinct from one another? [1]

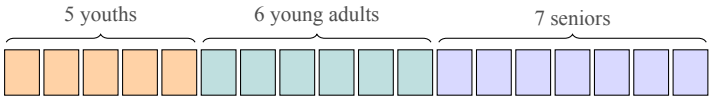
(ii) Find the total number of access codes that can be created given that there is no restriction. [3]

To run an event, the librarian plans to select a team of 12 volunteers from a group of 18. The volunteers comprise three age groups: 5 youths, 6 young adults, and 7 seniors.

(iii) In how many ways can the team be formed if at least one volunteer must be chosen from each age group? [3]

[ASRJC 2025 Prelim P2 Q11]

Solution		Comments
(a)(i) **[1]**	$\mathrm{P}(A \cup B') = \mathrm{P}(A)$ $= \dfrac{15+25+20+(18-n)}{100}$ or $\dfrac{100-22-n}{100}$ $= \dfrac{78-n}{100}$	
(a)(ii) **[2]**	$\mathrm{P}(F \mid A') = \dfrac{\mathrm{P}(F \cap A')}{1-P(A)} = \dfrac{\frac{n}{100}}{1-\frac{78-n}{100}} = \dfrac{n}{22+n}$	
(a)(iii) **[3]**	Given $\mathrm{P}(A \cap B) = \dfrac{3}{10}$ $\mathrm{P}(A \cap B) = \dfrac{25+18-n}{100} = \dfrac{43-n}{100}$ $\dfrac{43-n}{100} = \dfrac{3}{10} \Rightarrow n = 13$ $\mathrm{P}(A)\mathrm{P}(B) = \dfrac{65}{100} \times \dfrac{65}{100} = \dfrac{169}{400} \neq \dfrac{3}{10}$ Since $\mathrm{P}(A)\mathrm{P}(B) \neq \mathrm{P}(A \cap B)$, A and B are **NOT** independent **Alternatively**, comparing $\mathrm{P}(A \mid B)$ with $\mathrm{P}(A)$ $\mathrm{P}(A \mid B) = \dfrac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \dfrac{\frac{3}{10}}{\frac{65}{100}} = \dfrac{6}{13}$ Since $\mathrm{P}(A) = \dfrac{13}{20}$ and $\mathrm{P}(A \mid B) \neq \mathrm{P}(A)$, A and B are **NOT** independent	
(b)(i) **[1]**	3Es, 2Os, 2Ks, 1P, 1R, 1B Number of 4-letter code words $= {}^6C_4 \times 4! = 360$ or ${}^6P_4 = 360$	
(b)(ii) **[3]**	Case 1: All distinct (from b(i)) Number of 4-letter code words $= {}^6C_4 \times 4! = 360$ Case 2: One pair of repeated letters. (Eg XXYZ) Number of 4-letter code words $= {}^3C_1 \times {}^5C_2 \times \dfrac{4!}{2!} = 360$ Case 3: Two pairs of repeated letters. (Eg XXYY) Number of 4-letter code words $= {}^3C_2 \times \dfrac{4!}{2!2!} = 18$ Case 4: One set of 3 repeated letters (Eg EEEX) Number of 4-letter code words $= {}^5C_1 \times \dfrac{4!}{3!} = 20$ $\therefore$ Total number of 4-letter code word $= 360+360+18+20 = 758$ E E E O O K K P R B — 3 E, 2 O, 2 K (repeatable); P, R, B X Y Z W — Case 1: all distinct X X Y Z — Case 2: one pair X X Y Y — Case 3: two pairs X X X Y — Case 4: three alike (only E)	

(b)(iii) **[3]**	Total number of ways $= {}^{18}C_{12} = 18564$ No youth $= {}^{13}C_{12} = 13$ No young adult $= {}^{12}C_{12} = 1$ Total possible ways $= 18564 - 13 - 1 = 18550$  all teams – (no youth: 12 from 13) – (no young adult: 12 from 12); no senior is impossible (only 11 others)	

1.3 A group of 12 players is available for a street football match. The group consists of 5 attackers, 4 defenders and 3 goalkeepers. A team of 5 players is selected at random from the group.

(a) Find the

(i) number of different teams containing exactly 2 attackers, 2 defenders and 1 goalkeeper, [2]

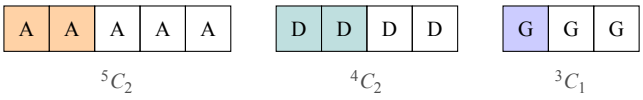
(ii) probability that the team contains exactly 1 goalkeeper and at most 2 attackers. [3]

The coach has already decided which goalkeeper will play in a match. To help him select from the 5 attackers and 4 defenders, he conducts a passing drill involving these players. For each pass, 1 player is chosen at random to receive the ball. After each pass, the ball is returned to the coach before the next player is chosen. A player may receive the ball more than once. The coach makes 4 passes in this manner. Find the probability that

(b) exactly 3 passes are made to attackers, [2]

(c) either the first 3 passes or the last 3 passes are made to attackers (or both). [3]

[JPJC 2026 Prelim P2 Q8]

Solution		**Comments**
(a)(i) **[2]**	Number of teams $= {}^5C_2 \times {}^4C_2 \times {}^3C_1 = 180$ 	

(a)(ii) **[3]**

Case 1: 1 goalkeeper, 2 attackers, 2 defenders $^3C_1 \times {}^5C_2 \times {}^4C_2 = 180$

Case 2: 1 goalkeeper, 1 attacker, 3 defenders $^3C_1 \times {}^5C_1 \times {}^4C_3 = 60$

Case 3: 1 goalkeeper, 0 attacker, 4 defenders $^3C_1 \times {}^5C_0 \times {}^4C_4 = 3$

Total number of ways $= {}^{12}C_5 = 792$

Hence probability that team has at most 2 attackers

$= \frac{180+60+3}{792} = \frac{243}{792}$ or 0.307

Case	G (3)	A (5)	D (4)	ways
1	1	2	2	180
2	1	1	3	60
3	1	0	4	3

team of 5: exactly 1 goalkeeper, at most 2 attackers

(b) **[2]**

Req probability

$= \mathrm{P}(\mathrm{AAAD}) + \mathrm{P}(\mathrm{AADA}) + \mathrm{P}(\mathrm{ADAA}) + \mathrm{P}(\mathrm{DAAA})$

$= \left(\frac{5}{9}\right)^3 \left(\frac{4}{9}\right) \times 4 = \frac{2000}{6561}$ or 0.305

pass 1	pass 2	pass 3	pass 4	
D	A	A	A	$\left(\frac{5}{9}\right)^3 \left(\frac{4}{9}\right)$
A	D	A	A	$\left(\frac{5}{9}\right)^3 \left(\frac{4}{9}\right)$
A	A	D	A	$\left(\frac{5}{9}\right)^3 \left(\frac{4}{9}\right)$
A	A	A	D	$\left(\frac{5}{9}\right)^3 \left(\frac{4}{9}\right)$

×4 positions for D

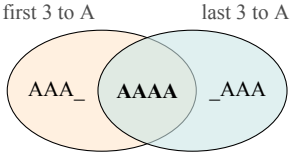
(c) **[3]**

<u>Method 1</u>:

Req probability $= \mathrm{P}(\mathrm{AAA_}) + \mathrm{P}(\mathrm{_AAA}) - \mathrm{P}(\mathrm{AAAA})$

$= \left(\frac{5}{9}\right)^3 + \left(\frac{5}{9}\right)^3 - \left(\frac{5}{9}\right)^4$

$\frac{1625}{6561}$ or 0.248



Method 1: add both, subtract overlap

Method 2: AAAD + DAAA + AAAA

<u>Method 2:</u>

Req prob $= \mathrm{P}(\mathrm{AAAD}) + \mathrm{P}(\mathrm{DAAA}) + \mathrm{P}(\mathrm{AAAA})$

$= \left(\frac{5}{9}\right)^3 \left(\frac{4}{9}\right) + \left(\frac{4}{9}\right) \left(\frac{5}{9}\right)^3 + \left(\frac{5}{9}\right)^4$

$\frac{1625}{6561}$ or 0.248

1.4 An office team of 10 people includes 7 men and 3 women named Anne, Beth, and Cathie. For an upcoming fire drill exercise, 5 individuals will be chosen, each assigned a unique role, to carry out the drill. Determine the number of possible ways to select 5 people from this group of 10

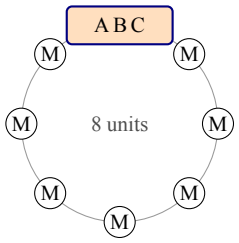
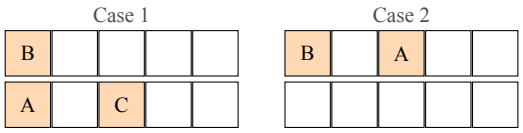
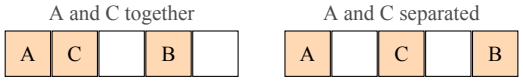
(i) to conduct the fire drill, [1]

(ii) such that at most 1 woman is selected to conduct the fire drill. [2]

(iii) After the fire drill exercise, the 10 people are to hold a discussion at a round table with 10 identical seats. Determine the number of ways in which Beth is seated between Anne and Cathie. [1]

(iv) A group photo of the 10 people, arranged in two rows of five, was taken after the discussion. Determine the number of ways in which Beth is not standing beside Anne or Cathie. [4]

[ASRJC 2024 Prelim P2 Q8]

Solution		Comments
(i) **[1]**	Number of ways $= \binom{10}{5} \times 5! = 30240$	
	role 1 | role 2 | role 3 | role 4 | role 5	
	choose 5 of 10: $\binom{10}{5}$, then assign the 5 distinct roles: 5!	
(ii) **[2]**	Number of ways $= \left[\binom{7}{5}\binom{3}{0} + \binom{7}{4}\binom{3}{1}\right] 5!$	
	$= (21 \times 1 + 35 \times 3)5! = 15120$	
(iii) **[1]**	Number of ways $= (8-1)!2! = 10080$	
	 $(8-1)!$ round the table, $2!$ for A B C or C B A	
(iv) **[4]**	Case 1 – Beth in 1 row while Anne and Cathie are in another row	
	$\binom{7}{4} \times 5! \times 5! \times 2 = 1008000$	
	Case 2 – Beth and one of them in 1 row	
	$\binom{7}{3} \times 3! \times \binom{2}{1} \times \binom{4}{2} \times 2! \times 5! \times 2 = 1209600$	
	Case 3 – Anne, Beth and Cathie are in the same row	
	$\underbrace{\binom{7}{2} \times 2! \times \binom{3}{2} \times 2! \times 2! \times 5! \times 2}_{\text{A and C together}} + \underbrace{\binom{7}{2} \times 2! \times 3! \times 5! \times 2}_{\text{A and C separated}} = 181440$	
	Thus number of ways $= 1008000 + 1209600 + 181440 = 2399040$	
	 Case 1: B in one row, A and C in the other. Case 2: B and one of A, C in the same row, not side by side.	
	 Case 3: A, B, C in the same row, B beside neither A nor C.	

2 Probability

2.1 An online game consists of three stages. All players start with Stage 1, and no matter whether they win or lose, they proceed to Stage 2. Players are allowed to proceed to Stage 3 only if they have won at least one of the first two stages.

Ming's probability of winning Stage 1 is p.

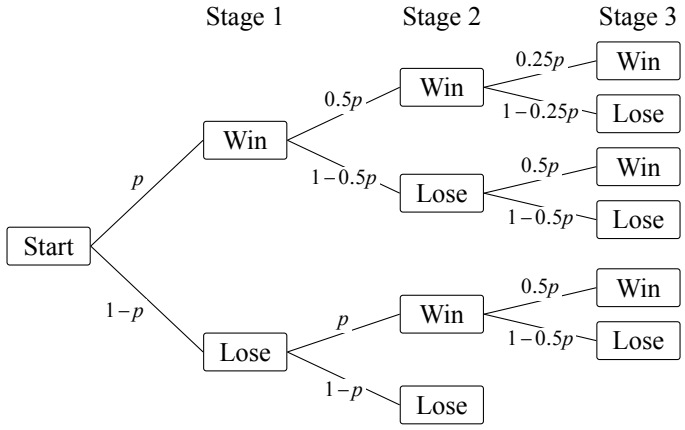
If Ming wins a stage, his probability of winning the next stage is halved. If he loses a stage, his probability of winning remains unchanged in the next stage.

(a) Draw a probability tree diagram to represent all possible outcomes for Ming over the three stages. [3]

(b) Find the range of possible values of p if the probability that Ming wins all three stages is more than 0.1. [2]

(c) It is now known that $p = 0.9$. Given that Ming proceeds to Stage 3, find the probability that he wins exactly two stages out of the three. [4]

[EJC 2025 Prelim P2 Q6]

Solution		Comments
(a) **[3]**		
(b) **[2]**	$p(0.5p)\ (0.5^2p) > 0.1$ $p^3 > 0.8$ $p > 0.928$	
(c) **[4]**	$\mathrm{P}(\text{wins exactly 2 stages out of 3} \mid \text{proceeds to Stage 3})$ $= \dfrac{\mathrm{P}(\text{wins exactly 2 stages out of 3} \cap \text{proceeds to Stage 3})}{\mathrm{P}(\text{proceeds to Stage 3})}$ $= \dfrac{0.9(0.45)(1-0.225)+0.9(0.55)(0.45)+0.1(0.9)(0.45)}{1-(0.1)(0.1)}$ $= \dfrac{0.577125}{0.99} = 0.583$	

2.2 (a) In a certain university, there are 110 Co-Curricular Activities (CCAs) clustered into 4 categories. There are 22 Arts and Culture CCAs, 16 Community Service CCAs, 38 Physical Sports CCAs and 34 Special Interest CCAs.

(i) Albert wishes to find out about approaches to training of the Physical Sports CCAs, so he sends a questionnaire to 22 Physical Sports CCAs. Explain whether these 22 Physical Sports CCAs form a sample or a population. [1]

(ii) Benedict wishes to investigate the level of student engagement in CCAs, but does not want to obtain the detailed information necessary from all 110 CCAs. Explain how he should carry out his investigation, and why he should do the investigation in this way. [2]

(iii) Find the number of different possible samples of 16 CCAs, with 4 CCAs chosen from each category. [2]

(b) The probability of events A, B and C occurring are given by $\mathrm{P}(A)=\frac{8}{25}$, $\mathrm{P}(B)=\frac{61}{100}$ and $\mathrm{P}(C)=\frac{7}{20}$ respectively. It is also known that $\mathrm{P}(A\cap C)=\frac{11}{100}$, $\mathrm{P}(B\mid C)=\frac{13}{35}$ and $\mathrm{P}(A\cap B\cap C)=\frac{7}{100}$.

(i) Find $\mathrm{P}(B\cap C)$. [2]

(ii) If events A and B are independent, find $\mathrm{P}(A\cap B\cap C')$. [2]

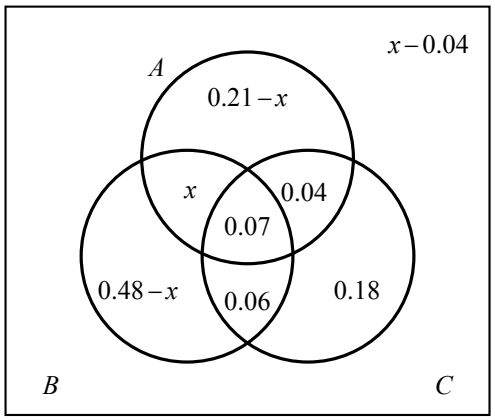
(iii) If events A and B are **not** known to be independent, find the greatest and least possible values of $\mathrm{P}(A\cap B\cap C')$. [4]

[CJC 2024 Prelim P2 Q10]

Solution		Comments
(a)(i) **[1]**	These 22 CCAs form a sample since this number is less than the population size of Physical Sports CCAs which is 38.	
(a)(ii) **[2]**	Benedict should select a random sample from the 110 CCAs: every CCA has an equal probability to be selected to be in the sample and the CCAS are selected independently. (If define without stating "random", need 2 full points.) A random sample helps to avoid bias.	
(a)(iii) **[2]**	Number of samples $={}^{22}C_4\times{}^{16}C_4\times{}^{38}C_4\times{}^{34}C_4$ $=4.56\times10^{16}$ (3 sf)	

(b)(i) **[2]**	$\mathrm{P}(B \mid C) = \frac{13}{35}$ $\frac{\mathrm{P}(B \cap C)}{\mathrm{P}(C)} = \frac{13}{35}$ $\frac{\mathrm{P}(B \cap C)}{\frac{7}{20}} = \frac{13}{35}$ $\mathrm{P}(B \cap C) = \frac{13}{100}$	
(b)(ii) **[2]**	$\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$ $= \frac{8}{25} \times \frac{61}{100}$ $= \frac{122}{625}$ $\mathrm{P}(A \cap B \cap C')$ $= \frac{122}{625} - \frac{7}{100}$ $= \frac{313}{2500}$	

(b)(iii)
[4]



Let $\mathrm{P}(A\cap B\cap C')=x$ where $0\leqslant x\leqslant 1$ ... (1)

Then

$$\mathrm{P}(A\cap B'\cap C')=\frac{8}{25}-\frac{11}{100}-x=\frac{21}{100}-x$$

$$\mathrm{P}(A'\cap B\cap C')=\frac{61}{100}-\frac{13}{100}-x=\frac{12}{25}-x$$

$\mathrm{P}(A\cup B\cup C)$

$$=\mathrm{P}(C)+\mathrm{P}(A\cap B'\cap C')+\mathrm{P}(A'\cap B\cap C')+\mathrm{P}(A\cap B\cap C')$$

$$=\frac{7}{20}+\left(\frac{21}{100}-x\right)+\left(\frac{12}{25}-x\right)+x$$

$$=\frac{26}{25}-x$$

Consider the probabilities $\mathrm{P}(A\cap B'\cap C')$, $\mathrm{P}(A'\cap B\cap C')$ and $\mathrm{P}(A\cup B\cup C)$,

$$0\leqslant \frac{21}{100}-x\leqslant 1 \quad \text{and} \quad 0\leqslant \frac{12}{25}-x\leqslant 1 \quad \text{and} \quad 0\leqslant \frac{26}{25}-x\leqslant 1$$

$$-\frac{79}{100}\leqslant x\leqslant \frac{21}{100} \text{ ... (2)} \qquad -\frac{13}{25}\leqslant x\leqslant \frac{12}{25} \text{ ... (3)}$$

$$\frac{1}{25}\leqslant x\leqslant \frac{26}{25} \text{ ... (4)}$$

Taking intersection of inequalities (1) to (4), $\frac{1}{25}\leqslant x\leqslant \frac{21}{100}$

$\min \mathrm{P}(A\cap B\cap C')=\frac{1}{25}$ and $\max \mathrm{P}(A\cap B\cap C')=\frac{21}{100}$

2.3 A bag contains one green disc, r red discs and $2r$ blue discs, where $r > 1$.

(a) Tim and Sim play a game by taking turns to draw a disc from the bag at random. The drawn disc is replaced after each turn. The first person to draw the green disc wins. If Sim started the game first, show that the probability of Sim winning the game is $\frac{3r+1}{6r+1}$. [2]

(b) In another game, a player draws 2 discs from the bag at random. If he draws a red disc, he scores 5 points, if he draws a blue disc, he scores 2 points and if he draws a green disc he scores 0. The player's score, denoted by T, is found by taking the difference in the number of points he scores for the two discs.

(i) Find $\mathrm{P}(T=t)$ for all possible values of t, leaving your answers in terms of r. [3]

(ii) Show that the expectation of the player's score is $\frac{2(2r+3)}{3r+1}$. [1]

(c) In a third game, a player draws a disc from the bag at random. If he draws a green disc, he wins \$0.15, if he draws a blue disc, he wins \$0.10 and if he draws a red disc, he loses \$0.25. May and Jun each draws a disc from the bag once, with the disc replaced in the bag after the first player drew. The probability that May ends up with more money than Jun is $\frac{27}{112}$. Find the value of r. [2]

Is the game fair for a player? Explain. [1]

[ACJC 2023 Prelim P2 Q9]

Solution		Comments
(a) **[2]**	$\frac{1}{3r+1}+\left(\frac{3r}{3r+1}\right)^2\times\frac{1}{3r+1}+\left(\frac{3r}{3r+1}\right)^4\times\frac{1}{3r+1}+\ldots$ $=\frac{\frac{1}{3r+1}}{1-\left(\frac{3r}{3r+1}\right)^2}=\frac{3r+1}{6r+1}$	
(b)(i) **[3]**	$\mathrm{P}(T=0)=\frac{r}{3r+1}\times\frac{r-1}{3r}+\frac{2r}{3r+1}\times\frac{2r-1}{3r}=\frac{5r-3}{3(3r+1)}$ $\mathrm{P}(T=2)=\frac{2r}{3r+1}\times\frac{1}{3r}\times 2!=\frac{4}{3(3r+1)}$ $\mathrm{P}(T=3)=\frac{2r}{3r+1}\times\frac{r}{3r}\times 2!=\frac{4r}{3(3r+1)}$ $\mathrm{P}(T=5)=\frac{r}{3r+1}\times\frac{1}{3r}\times 2!=\frac{2}{3(3r+1)}$ t | 0 | 2 | 3 | 5 $\mathrm{P}(T=t)$ | $\frac{5r-3}{3(3r+1)}$ | $\frac{4}{3(3r+1)}$ | $\frac{4r}{3(3r+1)}$ | $\frac{2}{3(3r+1)}$	
(b)(ii) **[1]**	$\mathrm{E}(T)=\sum t\mathrm{P}(T=t)$ $=0\times\frac{10r-4}{3(3r+1)}+2\times\frac{4}{3(3r+1)}+3\times\frac{4r}{3(3r+1)}+5\times\frac{2}{3(3r+1)}$ $=\frac{8+12r+10}{3(3r+1)}=\frac{2(2r+3)}{3r+1}$	
(c) **[1]**	x | -0.25 | 0.10 | 0.15 $\mathrm{P}(X=x)$ | $\frac{r}{3r+1}$ | $\frac{2r}{3r+1}$ | $\frac{1}{3r+1}$ $\mathrm{P}(X_1>X_2)=\mathrm{P}(X_1=0.15,X_2=-0.25)$ $+\mathrm{P}(X_1=0.15,X_2=0.10)$ $+\mathrm{P}(X_1=0.10,X_2=-0.25)$ $\frac{2r^2+3r}{(3r+1)^2}=\frac{27}{112}$ $19r^2-174r+27=0$ $r=\frac{3}{19}\approx 0.15789$ (rejected) $\quad\therefore r=9$ $\mathrm{E}(X)=\sum x\mathrm{P}(X=x)$ $=-0.25\times\frac{9}{28}+0.10\times\frac{18}{28}+0.15\times\frac{1}{28}$ $=-0.010714\approx-0.0107$ Since $\mathrm{E}(x)<0$, the game is <u>not</u> fair for a player, as he/she will be expected to lose 1.07 cents every time he/she plays.	

2.4 A basketball free throw game involves a team of two players, Ben and John. The game consists of at most three throws and the moment a player makes a successful shot, the team wins, and the game will end.

The game uses the following rules.

- Only one player is selected for each game and the probability that Ben is selected in a game is 0.7.
- The probabilities that Ben and John make a successful shot in any single attempt are 0.1 and 0.07 respectively.
- The shots are independent of each other.

(a) Find the probability that the team wins the game. [3]

(b) The team did not win the game. Find the probability that John was the one who was selected to shoot. [3]

(c) The team attempts the game repeatedly until the first game is won. Find the least number of attempts required such that the probability of winning within n games is at least 0.95. [2]

[HCI 2025 Prelim P2 Q6]

Solution		Comments
(a) **[3]**	$\mathrm{P}(\text{wins})$ $=\mathrm{P}(\text{wins and Ben selected})+\mathrm{P}(\text{wins and John selected})$ $=\mathrm{P}(\text{wins}\mid\text{Ben selected})\mathrm{P}(\text{Ben selected})$ $+\mathrm{P}(\text{wins}\mid\text{John selected})\mathrm{P}(\text{John selected})$ $=(0.1+0.9\times0.1+0.9^2\times0.1)\ (0.7)$ $+(0.07+0.93\times0.07+0.93^2\times0.07)\ (0.3)$ $=0.2483929$ OR $\mathrm{P}(\text{wins})$ $=\mathrm{P}(\text{wins and Ben selected})+\mathrm{P}(\text{wins and John selected})$ $=\mathrm{P}(\text{wins}\mid\text{Ben selected})\mathrm{P}(\text{Ben selected})+$ $\mathrm{P}(\text{wins}\mid\text{John selected})\mathrm{P}(\text{John selected})$ $=(1-0.9^3)\ (0.7)+(1-0.93^3)\ (0.3)$ $=0.2483929$	

(b) **[3]**

$$\text{P(John selected} \mid \text{did not win)}$$
$$= \frac{\text{P(John selected and did not win)}}{\text{P(did not win)}}$$
$$= \frac{0.3 \times 0.93^3}{1 - 0.2483929}$$
$$= 0.321 \text{ (3sf)}$$

(c) **[2]**

Required probability $= 1 -$ P(did not win in all n games)

$= 1 - (1 - 0.2483929)^n$

OR

Required probability

= P(win in 1st game) + P(win in 2nd game)

+ P(win in 3rd game) + … + P(win in n^{th} game)

$= 0.2483929 + (1 - 0.2483929)(0.2483929)$

$+ (1 - 0.2483929)^2 (0.2483929) + \ldots$

$+ (1 - 0.2483929)^{n-1} (0.2483929)$

$$= \frac{0.2483929\,(1 - (1 - 0.2483929)^n)}{1 - (1 - 0.2483929)}$$
$$= 1 - (1 - 0.2483929)^n$$

$1 - (1 - 0.2483929)^n \geqslant 0.95$

$(1 - 0.2483929)^n \leqslant 0.05$

n	$(1 - 0.2483929)^n$
10	$0.0575 > 0.05$
11	$0.0432 < 0.05$

Smallest n is 11.

See also Example 3 (b), (c) for another contextual probability question.

3 Discrete Random Variables

3.1 The probability distribution of a discrete random variable X is given as follows:

x	-2	-1	0	1	3
$\mathrm{P}(X=x)$	0.1	a	0.3	b	c

Given that $\mathrm{E}(X)=0.9$ and $\mathrm{Var}(X)=2.99$, find the values of a, b and c. [5]

[ACJC 2025 Prelim P2 Q8]

Solution		**Comments**
[5]	$\mathrm{E}(X)=(-2)(\frac{1}{10})+(-1)(a)+(0)(\frac{3}{10})+(1)(b)+(3)(c)$ $\frac{9}{10}=-\frac{2}{10}-a+b+3c$ $-a+b+3c=\frac{11}{10}$ ..... (1) $\mathrm{E}(X^2)=(-2)^2(\frac{1}{10})+(-1)^2(a)+(0)^2(\frac{3}{10})+(1)^2(b)+(3)^2(c)=$ $\frac{4}{10}+a+b+9c$ $\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2$ $\frac{299}{100}=\left(\frac{4}{10}+a+b+9c\right)-\left(\frac{9}{10}\right)^2$ $a+b+9c=\frac{34}{10}$ ..... (2) $\sum \mathrm{P}(X=x)=1$ $\frac{1}{10}+a+\frac{3}{10}+b+c=1$ $a+b+c=\frac{6}{10}$ ..... (3) Solving (1), (2) and (3), we have $a=\frac{1}{10}$ or 0.1, $b=\frac{3}{20}$ or 0.15, $c=\frac{7}{20}$ or 0.35	

3.2 Anthony plays a game using a four-sided fair die with the numbers 1, 2, 3 and 4 printed on its sides. Anthony rolls the die and should the outcome be a prime number, he wins an amount equivalent to that number in dollars. Conversely, if the outcome is a non-prime number, Anthony will lose an amount equivalent to that number in dollars. The random variable X represents the amount of money Anthony wins in a roll of the die.

(a) Tabulate the probability distribution of X and find the value of $\mathrm{E}(X)$. [2]

(b) Find the value of the variance of X. [1]

(c) The random variables X_1 and X_2 are two independent observations of X. By drawing a table of outcomes for $|X_1+X_2|$ or otherwise, find $\mathrm{E}(|X_1+X_2|)$. [2]

Assume now that a biased six-sided die with numbers 2, 3, 4, 5, 6 and 7 printed on its sides, is used instead for the same game. It is known that for this biased die, the probability of obtaining a "3" is p, the probability of obtaining a "5" is $2p$ and the remaining four numbers each have an equal probability of occurring.

(d) Find the exact value of p for the game to be fair using the biased die. [3]

[ASRJC 2025 Prelim P2 Q7]

Solution		Comments
(a) **[2]**	$\begin{array}{\|c\|c\|c\|c\|c\|}\hline x & -4 & -1 & 2 & 3 \\ \hline \mathrm{P}(X=x) & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \hline \end{array}$ $\mathrm{E}(X)=\frac{-4}{4}+\frac{-1}{4}+\frac{2}{4}+\frac{3}{4}=0$	
(b) **[1]**	$\mathrm{Var}(X)=\left[\frac{(-4)^2}{4}+\frac{(-1)^2}{4}+\frac{2^2}{4}+\frac{3^2}{4}\right]-0^2=\frac{15}{2}$	
(c) **[2]**	Let $Y=\lvert X_1+X_2\rvert$ $\begin{array}{\|c\|c\|c\|c\|c\|}\hline X_2 \backslash X_1 & -4 & -1 & 2 & 3 \\ \hline -4 & 8 & 5 & 2 & 1 \\ \hline -1 & 5 & 2 & 1 & 2 \\ \hline 2 & 2 & 1 & 4 & 5 \\ \hline 3 & 1 & 2 & 5 & 6 \\ \hline \end{array}$ $\begin{array}{\|c\|c\|c\|c\|c\|c\|c\|}\hline y & 1 & 2 & 4 & 5 & 6 & 8 \\ \hline \mathrm{P}(Y=y) & \frac{4}{16} & \frac{5}{16} & \frac{1}{16} & \frac{4}{16} & \frac{1}{16} & \frac{1}{16} \\ \hline \end{array}$ $\mathrm{E}(Y)=\frac{4}{16}+2\left(\frac{5}{16}\right)+4\left(\frac{1}{16}\right)+5\left(\frac{4}{16}\right)+6\left(\frac{1}{16}\right)+8\left(\frac{1}{16}\right)=\frac{13}{4}$	
(d) **[3]**	The new probability distribution for X is as tabulated below: $\begin{array}{\|c\|c\|c\|c\|c\|c\|c\|}\hline x & -6 & -4 & 2 & 3 & 5 & 7 \\ \hline \mathrm{P}(X=x) & \frac{1-3p}{4} & \frac{1-3p}{4} & \frac{1-3p}{4} & p & 2p & \frac{1-3p}{4} \\ \hline \end{array}$ $\mathrm{E}(X)=-6\left(\frac{1-3p}{4}\right)-4\left(\frac{1-3p}{4}\right)+2\left(\frac{1-3p}{4}\right)+3p+10p+$ $7\left(\frac{1-3p}{4}\right)=-\left(\frac{1-3p}{4}\right)+13p=-\frac{1}{4}+\frac{55}{4}p$ For the game to be fair, $-\frac{1}{4}+\frac{55}{4}p=0$ Hence $p=\frac{1}{55}$	

3.3 Andrea participates in a game show. In each round of the game, Andrea responds to a series of questions until she answers a question wrongly or she has answered five questions correctly. She is awarded one point for every correct answer and is awarded an additional 3 bonus points if she answers all five questions correctly. It is given that she answers each question independently and the probability she answers each question correctly is $\frac{4}{5}$.

(a) Show that the probability that Andrea obtains 2 points in one round of the game is $\frac{16}{125}$. [1]

(b) Find the probability distribution of the points that Andrea obtains in one round of the game. [2]

(c) Find the expectation and variance of the points that Andrea obtains in one round of the game. [2]

(d) Find the probability that Andrea obtains less than 3 points after two rounds of the game. [3]

[CJC 2024 Prelim P2 Q5]

Solution		Comments
(a) **[1]**	Probability that Andrea obtains 2 points in a round of the game $=\left(\frac{4}{5}\right)\left(\frac{4}{5}\right)\left(\frac{1}{5}\right)=\frac{16}{125}$	
(b) **[2]**	Let X be the random variable denoting the number of points obtained in a round.	
(c) **[2]**	**Method 1**: Using G.C., Expectation $=3.67232$ (exact); Variance $=3.22334^2=10.4$ (3 s.f.)	
	Method 2: $\mathrm{E}(X)=0\left(\frac{1}{5}\right)+1\left(\frac{4}{25}\right)+2\left(\frac{16}{125}\right)+3\left(\frac{64}{625}\right)+4\left(\frac{256}{3125}\right)+8\left(\frac{1024}{3125}\right)=3.67232$ $\mathrm{E}(X^2)=0^2\left(\frac{1}{5}\right)+1^2\left(\frac{4}{25}\right)+2^2\left(\frac{16}{125}\right)+3^2\left(\frac{64}{625}\right)+4^2\left(\frac{256}{3125}\right)+8^2\left(\frac{1024}{3125}\right)=23.87584$ $\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2=23.87584-(3.67232)^2=10.4$ (to 3 s.f.)	
(d) **[3]**	Probability that Andrea obtains less than 3 points after two rounds $=\mathrm{P}(X_1+X_2<3)=\mathrm{P}(X_1+X_2=0)+\mathrm{P}(X_1+X_2=1)+\mathrm{P}(X_1+X_2=2)$ $=\mathrm{P}(X_1=0\cap X_2=0)+2\mathrm{P}(X_1=0\cap X_2=1)+2\mathrm{P}(X_1=0\cap X_2=2)+\mathrm{P}(X_1=1\cap X_2=1)$ $=\left(\frac{1}{5}\right)^2+2\left(\frac{1}{5}\right)\left(\frac{4}{25}\right)+2\left(\frac{1}{5}\right)\left(\frac{16}{125}\right)+\left(\frac{4}{25}\right)^2=\frac{113}{625}$ or 0.1808 (exact)	

x	0	1	2	3	4	8
$\mathrm{P}(X=x)$	$\frac{1}{5}$	$\frac{4}{25}$	$\frac{16}{125}$	$\frac{64}{625}$	$\frac{256}{3125}$	$\frac{1024}{3125}$

3.4 The number of muffins, X, bought by a randomly chosen customer at a muffin shop has probability distribution as shown in the table below:

x	1	2	3	4	5	6	>6
$\mathrm{P}(X=x)$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	a	$\frac{2}{12}$	$\frac{1}{12}$	0

(i) Find the value of a.

(ii) State the expected value of X.

(iii) Evaluate the variance of X.

Initially, the shopkeeper charges \$2.50 for each muffin, which cost 50 cents to produce.

(iv) Using the answer to part (ii), or otherwise, find the expected profit made per customer.

The shopkeeper decides to promote the sale of muffins by giving a free muffin to all customers who buy 4 or more muffins. As a result, the probability distribution of the number of muffins, Y, taken away by a customer is given in the following table.

y	1	2	3	4	5	6	>6
$\mathrm{P}(Y=y)$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	0	$\frac{20}{36}$	$\frac{6}{36}$	0

(v) Find the expected profit made per customer.

(vi) If the number of customers increases by 8%, is the promotion more profitable for the shopkeeper? Give a reason for your answer.

[SAJC DRV HW2 Review Q8]

Solution		Comments
(i)	$\frac{1}{4}$	
(ii)	3.5	
(iii)	$1\frac{11}{12}$	
(iv)	*No official answer is printed for this part.*	
(v)	\$6.86	
(vi)	*No official answer is printed for this part.*	

Only the answer line printed with the question (SAJC DRV HW2 Review Q8) is available; no official worked solution was found.

3.5 In a game, a player selects one ball at random from a bag which contains four balls numbered '1', '1', '2' and '3'. The player notes the number on the selected ball, and then tosses that number of fair coins. The number of heads obtained is denoted by X and has a probability distribution as follows:

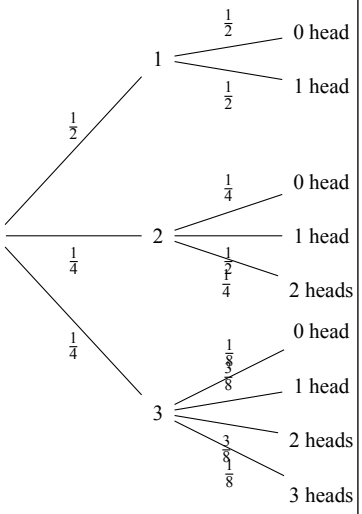
x	0	1	2	3
$\mathrm{P}(X=x)$	a	$\frac{15}{32}$	b	$\frac{1}{32}$

(i) Find the exact values of a and b. [3]

(ii) Find $\mathrm{E}(X)$ and show that $\mathrm{Var}(X)=\frac{39}{64}$. [2]

(iii) The player plays 50 games. Find the probability that the average number of heads obtained per game exceed 1. [2]

[JPJC 2025 Prelim P2 Q7]

Solution		Comments
(i) **[3]**	$a = \mathrm{P}(X=0) = \left(\frac{1}{2}\times\frac{1}{2}\right) + \left(\frac{1}{4}\times\left(\frac{1}{2}\right)^2\right) + \left(\frac{1}{4}\times\left(\frac{1}{2}\right)^3\right) = \frac{11}{32}$ $b = \mathrm{P}(X=2) = \left(\frac{1}{4}\times\left(\frac{1}{2}\right)^2\right) + \left(\frac{1}{4}\times{}^3C_2\times\left(\frac{1}{2}\right)^3\right) = \frac{5}{32}$ Or $b = \mathrm{P}(X=2) = 1 - \frac{11}{32} - \frac{15}{32} - \frac{1}{32} = \frac{5}{32}$ $a = \frac{11}{32}, b = \frac{5}{32}$ 	
(ii) **[2]**	Probability distribution of X: x: 0, 1, 2, 3 $\mathrm{P}(X=x)$: $\frac{11}{32}$, $\frac{15}{32}$, $\frac{5}{32}$, $\frac{1}{32}$ $\mathrm{E}(X) = \sum x\mathrm{P}(X=x) = (0)\left(\frac{11}{32}\right) + (1)\left(\frac{15}{32}\right) + (2)\left(\frac{5}{32}\right) + (3)\left(\frac{1}{32}\right) = \frac{7}{8}$ $\mathrm{E}(X^2) = \sum x^2\mathrm{P}(X=x) = (0)^2\left(\frac{11}{32}\right) + (1)^2\left(\frac{15}{32}\right) + (2)^2\left(\frac{5}{32}\right) + (3)^2\left(\frac{1}{32}\right) = \frac{11}{8}$ $\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 = \frac{11}{8} - \left(\frac{7}{8}\right)^2 = \frac{39}{64}$ (Shown)	
(iii) **[2]**	Let X = number of heads obtained in one game; $\overline{X}$ = mean number of heads obtained per game Since $n = 50$ is large, by the Central Limit Theorem, $\overline{X} \sim \mathrm{N}\left(\frac{7}{8}, \frac{39/64}{50}\right) = \mathrm{N}\left(\frac{7}{8}, \frac{39}{3200}\right)$ approximately $\mathrm{P}(\overline{X} > 1) = 0.12876 \approx 0.129$	

4 Binomial Distribution

4.1 A fruit seller sources his oranges from a supplier with a large plantation where 1% of the oranges are rotten. The supplier picks the oranges regardless of whether the oranges are rotten or not, and packs 20 oranges in each box at random.

(a) The random variable X denotes the number of rotten oranges in a randomly chosen box. State in context two assumptions for X to follow a binomial distribution. [2]

Assume that X follows a binomial distribution for the rest of this question.

The fruit seller considers a box containing between 2 and 5 rotten oranges inclusive as substandard.

(b) Find the probability that a randomly chosen box is substandard. [1]

(c) Find the expected number of rotten oranges in a box. [1]

A truckload of 10 randomly chosen boxes of oranges is then sent to the fruit seller.

(d) Find the probability that not more than 2 boxes from the entire truckload are substandard. [2]

Upon receiving the oranges, the fruit seller inspects the boxes with these conditions:

- Accept the entire truckload if the first box opened has no rotten oranges.
- Reject the entire truckload if the first box opened has more than 1 rotten orange.
- If the first box has exactly 1 rotten orange, the fruit seller opens a second box.
- If the second box has at most 1 rotten orange, he will accept the entire truckload. Otherwise, he will reject the entire truckload.

(e) Find the probability that he will reject the entire truckload. [2]

[ACJC 2025 Prelim P2 Q9]

Solution		Comments
(a) **[2]**	The probability that an orange is rotten is a constant of 0.01. Whether an orange is rotten is independent of whether any other oranges are rotten.	
(b) **[1]**	Let the random variable X denote the number of rotten oranges in a box of 20 oranges. $X \sim \mathrm{B}(20, 0.01)$ $\mathrm{P}(2 \leqslant X \leqslant 5) = \mathrm{P}(X \leqslant 5) - \mathrm{P}(X \leqslant 1) = 0.0169$	
(c) **[1]**	$\mathrm{E}(X) = (20)(0.01) = 0.2$	
(d) **[2]**	Let the random variable Y denote the number of substandard boxes out of the entire truckload of 10 boxes. $Y \sim \mathrm{B}(10, p)$ where $p = 0.0169$ $\mathrm{P}(Y \leqslant 2) = 0.999$ (3 s.f.)	
(e) **[2]**	Let A = {he rejects the entire truckload} $\mathrm{P}(A) = \mathrm{P}(X \geqslant 2) + \mathrm{P}(X = 1)\mathrm{P}(X \geqslant 2) = [1 - \mathrm{P}(X \leqslant 1)] + \mathrm{P}(X = 1)[1 - \mathrm{P}(X \leqslant 1)] = 0.0196$	

4.2 A manufacturer produces a large number of tumblers of three colours: black, pink and white in the respective ratios $3:2:1$. The tumblers are packed into boxes of 20 each.

Assume that the number of white tumblers in each box follows a binomial distribution.

(a) Find the probability that the number of white tumblers in a randomly chosen box is between 4 and 9 inclusive. [2]

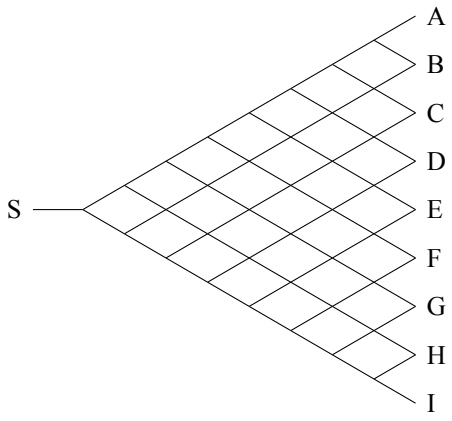
(b) Find the probability that the 19th tumbler chosen from the box is the fourth white tumbler. [2]

It is known that, on average, p% of the tumblers are defective. The number of defective tumblers in a box also follows a binomial distribution. The probability that a box contains at most one defective tumbler is 0.95.

(c) Write down an equation satisfied by p. Hence find the value of p. [3]

[YIJC 2024 Prelim P2 Q8]

Solution		Comments
(a) **[2]**	Let X be the number of white tumblers in a box of 20. $X \sim \mathrm{B}(20, \frac{1}{6})$ $\mathrm{P}(4 \leqslant X \leqslant 9) = \mathrm{P}(X \leqslant 9) - \mathrm{P}(X \leqslant 3) \approx 0.433$ Alternative: $\mathrm{P}(X=4) + \mathrm{P}(X=5) + \cdots + \mathrm{P}(X=9) \approx 0.433$	
(b) **[2]**	Let W be the number of white tumblers from 18 tumblers. $W \sim \mathrm{B}(18, \frac{1}{6})$ Required probability $= \mathrm{P}(W=3) \times \frac{1}{6} \approx 0.0409$	
(c) **[3]**	Let D be the number of defective tumblers in a box of 20. $D \sim \mathrm{B}(20, 0.01p)$ $\mathrm{P}(D \leqslant 1) = 0.95$ $\mathrm{P}(D=0) + \mathrm{P}(D=1) = 0.95$ $^{20}C_0 (0.01p)^0 (1-0.01p)^{20} + {}^{20}C_1 (0.01p)^1 (1-0.01p)^{19} = 0.95$ $(1-0.01p)^{20} + 0.2p(1-0.01p)^{19} = 0.95$ Using G.C., $p \approx 1.81$	



4.3

In a computer game, a bug moves from left to right through a network of connected paths. The bug starts at S and, at each junction, randomly takes the left fork with probability p or the right fork with probability q, where $q = 1 - p$. The forks taken at each junction are independent. The bug finishes its journey at one of the 9 endpoints labelled A–I (see diagram).

(i) Show that the probability that the bug finishes its journey at D is $56p^5q^3$. [2]

(ii) Given that the probability that the bug finishes its journey at D is greater than the probability that the bug finishes its journey at any one of the other endpoints, find exactly the possible range of values of p. [4]

In another version of the game, the probability that, at each junction, the bug takes the left fork is $0.9p$, the probability that the bug takes the right fork is $0.9q$ and the probability that the bug is swallowed up by a 'black hole' is 0.1.

(iii) Find the probability that, in this version of the game, the bug reaches one of the end-points A–I, without being swallowed up by a black hole. [1]

[N2018 P2 Q6]

Solution		**Comments**
(i) **[2]**	*No official answer is printed for this part.*	
(ii) **[4]**	$\frac{5}{9} < p < \frac{2}{3}$	
(iii) **[1]**	0.430	

No official SEAB solution is published; answers as printed with this question in RI 2025 Tutorial S2B Q10.

4.4 'Crunchers' are sweets that are sold in packets of 8. Each packet is made up of randomly chosen coloured sweets. On average 15% of Crunchers are yellow.

(a) Find the probability that a randomly chosen packet of Crunchers contains no more than one yellow sweet. [2]

(b) Kev buys 70 randomly chosen packets of Crunchers. Find the probability that at least 50 of these packets contain no more than one yellow sweet. [3]

On average the proportion of Crunchers that are red is p. It is known that the modal number of red sweets in a packet is 3.

(c) Use this information to find exactly the range of values that p can take. [4]

[9758 Specimen Paper (for 2025) P2 Q7]

Solution		Comments
(a) **[2]**	Use of B(8, 0.15); 0.657	
(b) **[3]**	Use of B(70, 0.657); Use of $1-\mathrm{P}(x \leqslant 49)$; 0.189	
(c) **[4]**	$\frac{8\times7\times6}{1\times2\times3}p^3(1-p)^5 > \frac{8\times7}{1\times2}p^2(1-p)^6; \quad p > \frac{1}{3}$ $\frac{8\times7\times6}{1\times2\times3}p^3(1-p)^5 > \frac{8\times7\times6\times5}{1\times2\times3\times4}p^4(1-p)^4; \quad p < \frac{4}{9}$	

4.5 A factory produces porcelain bowls. It is known that, on average, 8% of the bowls are faulty. The bowls are packed in boxes of 20. A box is considered imperfect if it contains more than 2 faulty bowls. Assume that the number of faulty bowls in a box follows a binomial distribution.

(a) Find the probability that a randomly chosen box is imperfect. [2]

These boxes are packed into cartons of 12 boxes each.

(b) Find the probability that, out of 3 randomly chosen cartons, there are 2 cartons that contains fewer than 15 faulty bowls each. [3]

Let Y be the number of imperfect boxes in a randomly chosen carton.

(c) State the values of $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$. [2]

(d) Hence using a suitable approximation, find the probability that the total number of imperfect boxes in a random sample of 35 cartons is at most 85. [2]

[TJC 2025 Prelim P2 Q7]

Solution		Comments
(a) **[2]**	Let X be the number of faulty bowls in a box of 20 bowls. $X \sim \mathrm{B}(20, 0.08)$ $\mathrm{P}(X > 2) = 1 - \mathrm{P}(X \leqslant 2) = 1 - 0.78795 = 0.21205$ (5 s.f.) $= 0.212$ (3 s.f.)	
(b) **[3]**	Let W be the number of faulty bowls in a carton (of 12 boxes). $W \sim \mathrm{B}(12 \times 20, 0.08)$, i.e. $W \sim \mathrm{B}(240, 0.08)$ $\mathrm{P}(W < 15) = \mathrm{P}(W \leqslant 14) = 0.12933$ (5 s.f.) Required Probability $= {}^3C_2 (0.12933)^2 (1 - 0.12933) = 0.0437$ (3 s.f.) **OR** Let A be the number of cartons out of 3 contains fewer than 15 faulty bowls each. $A \sim \mathrm{B}(3, 0.12933)$ Required Probability $= \mathrm{P}(A = 2) = 0.0437$ (3 s.f.)	
(c) **[2]**	$Y \sim \mathrm{B}(12, 0.21205)$ $\mathrm{E}(Y) = np = 12 \times 0.21205 = 2.5446 = 2.54$ (3 s.f.) $\mathrm{Var}(Y) = np(1-p) = 12 \times 0.21205 \times (1 - 0.21205) = 2.0050$ (5 s.f.) $= 2.01$ (3s.f.)	
(d) **[2]**	$Y \sim \mathrm{B}(12, 0.21205)$. Let $T = Y_1 + \cdots + Y_{35}$. Since sample size 35 is large, by Central Limit Theorem, $T \sim \mathrm{N}(35 \times 2.5446,\ 35 \times 2.0050)$ approximately i.e. $T \sim \mathrm{N}(89.061,\ 70.175)$ approximately Required probability $= \mathrm{P}(T \leqslant 85) = 0.314$ (3 s.f.)	

5 Normal Distribution

5.1 In this question, you should state clearly all the distributions that you use, together with the values of the appropriate parameters.

A toy designer creates weighted wooden blocks used in balance-based educational toys. There are two types of blocks: cylindrical blocks and cuboidal blocks. The mass, in grams, of each cylindrical block and cuboidal block follow a normal distribution with the following parameters:

	Mean	Standard Deviation
cylindrical block	280	6
cuboidal block	220	5

(a) Find the probability that the mass of a randomly selected cylindrical block exceeds the mass of a randomly selected cuboidal block by more than 65 grams. [2]

(b) The probability that the total of twice the mass of a cylindrical block and thrice the mass of a cuboidal block being less than M g is 0.355. Find the value of M, giving your answer to the nearest integer. [2]

To build a deluxe seesaw model, the designer uses:

- 2 cylindrical blocks,
- 3 cuboidal blocks,
- 7 screws,
- 1 wooden plank.

The blocks are drilled before assembly to fit screws:

- Each cylindrical block loses 5% of its mass after drilling.
- Each cuboidal block loses 3% of its mass after drilling.

The masses of the remaining components follow a normal distribution with the following parameters:

	Mean	Standard Deviation
screw	8	1.2
wooden plank	540	8

When assembling the deluxe seesaw model, all of its components are chosen at random.

(c) Find the probability that the total mass of all the modified components in a deluxe seesaw model is more than 1785 g. [4]

[ASRJC 2025 Prelim P2 Q9]

Solution		Comments
(a) **[2]**	Let X and Y be the mass of a randomly selected cylindrical and cuboidal block (in g) respectively. $X \sim \mathrm{N}(280,\ 6^2)$ and $Y \sim \mathrm{N}(220,\ 5^2)$ $X-Y \sim \mathrm{N}(280-220,\ 5^2+6^2)$ $X-Y \sim \mathrm{N}(60,\ 61)$ $\mathrm{P}(X-Y>65)=0.261$	
(b) **[2]**	Let $C=2X+3Y$ $C \sim \mathrm{N}(2\times280+3\times220,\ 2^2\times6^2+3^2\times5^2)$ $C \sim \mathrm{N}(1220,\ 369)$ Given $\mathrm{P}(C<M)=0.355$ From GC, $M=1212.8568$ $M=1213$	
(c) **[4]**	Let S and W be the mass of a randomly selected screw and wooden plank (in g) respectively. $S \sim \mathrm{N}(8,\ 1.2^2)$ and $W \sim \mathrm{N}(540,\ 8^2)$ Let $T=0.95(X_1+X_2)+0.97(Y_1+Y_2+Y_3)+S_1+\cdots+S_7+W$ $\mathrm{E}(T)=(0.95\times2\times280)+(0.97\times3\times220)+(7\times8)+540$ $=532+426.8+56+540$ $=1768.2$ $\mathrm{Var}(T)=(0.95^2\times2\times6^2)+(0.97^2\times3\times5^2)+(7\times1.2^2)+64$ $=64.98+70.5675+10.08+64$ $=209.6275$ $\mathrm{P}(T>1785)=0.123$	

5.2 In this question you should state the parameters of any distribution you use.

An electric power service company keeps records of the installation time of its electricity meters in the houses in a new estate. The time taken to install an electricity meter is normally distributed with mean 45 minutes and standard deviation 6 minutes.

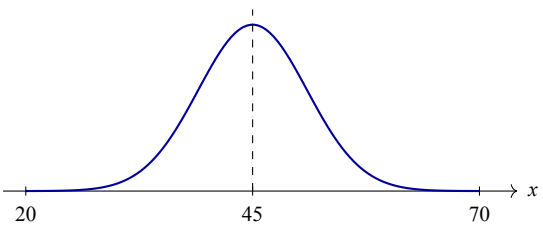
(a) Sketch the distribution for the installation time of an electricity meter between 20 minutes and 70 minutes. [2]

(b) A house that has its electricity meter take more than 1 hour to be installed is considered 'inefficient'. The company randomly selects n houses in the estate for quality control in their services. Find the greatest value of n such that the probability that fewer than 3 of these n houses are 'inefficient' is at least 0.90. [3]

(c) Each month, the amount of electricity, in kilowatt-hours (kWh), used by a particular household in the estate has the distribution $\mathrm{N}(524, 27^2)$. The company charges households for electricity used at \$0.26 per kWh and each household is billed every two months. Find the probability that a randomly chosen bill for this household is more than \$270 given that it is between \$250 and \$280. State an assumption that is needed for your calculations to be valid. [4]

(d) The company also installs gas meters in the houses in the estate. The time taken in minutes to install a gas meter follows a normal distribution. 40% of the gas meters each has an installation time greater than 53 minutes and 15% of the gas meters each has an installation time less than 38 minutes. Find the mean and standard deviation of the installation times of the gas meters in the estate. [4]

[NJC 2024 Prelim P2 Q12]

Solution		**Comments**
(a) **[2]**	Let X denote the random variable 'time taken (min) to install an electricity meter.' $X \sim \mathrm{N}(45, 6^2)$  Note: $\mathrm{P}(20 < X < 70) = 0.99997$	

(b) **[3]**

Let E be the random variable 'number of 'inefficient' houses out of n houses.'

$E \sim \mathrm{B}(n,\ \mathrm{P}(X>60))$

$E \sim \mathrm{B}(n,\ 0.0062097)$

$\mathrm{P}(E<3) = \mathrm{P}(E \le 2) \ge 0.90$

Using GC,

n	$\mathrm{P}(E \le 2)$
177	$0.9012 \ge 0.9$
178	$0.8999 < 0.9$
179	0.8986

Largest $n = 177$

(c) **[4]**

Let W be the random variable 'amount of electricity (kWh) used in the household in a month.'

$W \sim \mathrm{N}(524,\ 27^2)$

Let $T = 0.26(W_1 + W_2)$.

$\mathrm{E}(T) = 0.26 \times 524 \times 2 = 272.48$

$\mathrm{Var}(T) = 0.26^2 \times 2 \times 27^2 = 98.5608$

$T \sim \mathrm{N}(272.48,\ 98.5608)$

$$\mathrm{P}(T>270 \mid 250<T<280) = \frac{\mathrm{P}(270<T<280)}{\mathrm{P}(250<T<280)}$$
$$= \frac{0.36425}{0.76383}$$
$$= 0.490 \ \text{(3 s.f.)}$$

We need to assume that the electricity used in each of these two months is independent for this particular household.

(d) **[4]**

Let G denote the random variable 'time taken (min) to install a gas meter

$G \sim \mathrm{N}(\mu, \sigma^2)$.

$\mathrm{P}(G<38) = 0.15, \qquad \mathrm{P}(G>53) = 0.4$

$\mathrm{P}\left(Z < \frac{38-\mu}{\sigma}\right) = 0.15, \qquad \mathrm{P}\left(Z > \frac{53-\mu}{\sigma}\right) = 0.4$

By G.C.,

$\frac{38-\mu}{\sigma} = -1.0364, \qquad \frac{53-\mu}{\sigma} = 0.25335$

$\mu - 1.0364\sigma = 38 \quad ---(1), \qquad \mu + 0.25335\sigma = 53 \quad ---(2)$

Solving (1) and (2):

$\mu = 50.0535 = 50.1$ (3 s.f.)

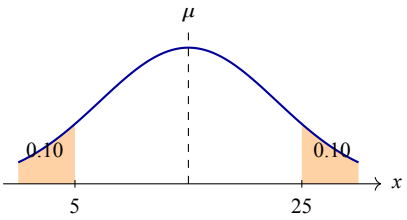
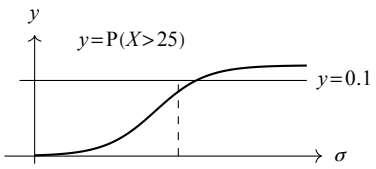
$\sigma = 11.6298 = 11.6$ (3 s.f.)

5.3 The waiting times for customers to get their pastries from a popular bakery is modelled by a normal distribution with mean μ minutes and standard deviation σ minutes. On average, 10% of customers wait for more than 25 minutes and the same percentage of customers wait for less than 5 minutes.

(i) Find the values of μ and σ, leaving your answers to the nearest minute. [3]

(ii) Explain if the normal distribution is a suitable model for the waiting times for customers to get pastries from this bakery. [1]

[MI 2025 Prelim P2 Q6]

Solution		**Comments**
(i) **[3]**	Let X be the waiting time to get pastries from the popular bakery of a randomly chosen customer. $X \sim \mathrm{N}(\mu,\ \sigma^2)$ Given $\mathrm{P}(X<5)=\mathrm{P}(X>25)=0.10$. **Method 1** By symmetry, $\mu=\dfrac{5+25}{2}=15$  $\mathrm{P}(X>25)=0.10$ $\mathrm{P}\left(Z>\dfrac{25-15}{\sigma}\right)=0.10$ $\dfrac{10}{\sigma}=1.28155$ $\sigma=7.80305$ --- ***Alternatively,*** $\mathrm{P}(X>25)=0.10$ Using GC,  $\sigma=7.80304$ --- $\therefore \mu=15,\ \sigma=8$ (to the nearest minute).	

	Method 2 $\mathrm{P}(X>25)=0.10$ $\quad$ $\mathrm{P}(X<5)=0.10$ $\mathrm{P}\left(Z>\frac{25-\mu}{\sigma}\right)=0.10$ $\quad$ $\mathrm{P}\left(Z<\frac{5-\mu}{\sigma}\right)=0.10$ $\frac{25-\mu}{\sigma}=1.28155$ $\quad$ $\frac{5-\mu}{\sigma}=-1.28155$ $\mu+1.28155\sigma=25$ … (1) $\quad$ $\mu-1.28155\sigma=5$ … (2) Solving (1) and (2) simultaneously, $\mu=15,\ \sigma=7.80305.$ $\therefore\mu=15,\ \sigma=8$ (to the nearest minute).	
(ii) **[1]**	Consider $X\sim\mathrm{N}(15,\ 8^2)$. $\mathrm{P}(X<0)=0.0303.$ (or 0.02728 if we use $\sigma=7.80305$) The waiting times cannot be negative but $\mathrm{P}(X<0)$ is about 3%, which is not insignificant. Thus, the normal distribution is not a suitable model. **Note:** For normal distribution: $\mathrm{P}(\mu-3\sigma<X<\mu-3\sigma)\approx 0.997$ i.e. 99.7%. Hence, whenever the probability of obtaining unreasonable values of x exceeds 0.003, we can conclude that the normal distribution is not suitable	

5.4 A factory produces metal discs and metal bars for constructing machinery parts. The mass of a randomly chosen metal disc follows a normal distribution with mean 18 grams and standard deviation 0.6 grams. The mass of a randomly chosen metal bar has an independent normal distribution with mean 20 grams and standard deviation σ grams.

(a) Sketch the distribution for the masses of the metal disc between 14 grams and 22 grams. [1]

(b) 2 metal discs are randomly chosen. Find the probability that the mass of one of the discs is less than 19 grams and the other metal disc has mass more than 19 grams. [2]

(c) Find the range of values of k such that less than 90% of the metal discs have masses within k grams of the mean mass of the metal discs. [2]

(d) It is known that 7.65% of the metal bars have masses more than 21 grams. Find the value of σ. [2]

It is given that $\sigma = 0.5$ for the rest of the question.

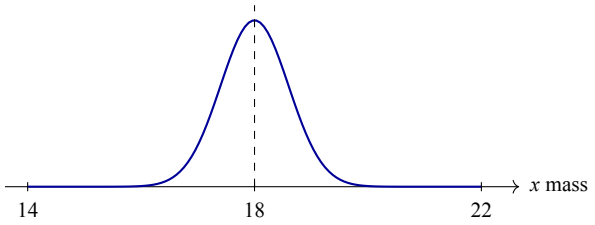
(e) A client places an order for a new kind of metal strips. The metal strips are modelled as similar solids to the metal bars with a 20% reduction in mass.

A production worker selects 8 of the metal strips for inspection. Find the probability that at least 5 of the metal strips have masses between 15.8 grams and 17.2 grams. [3]

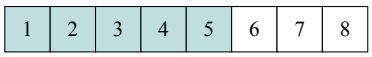
(f) A protective case is designed for the safe transportation of the metal bar. Each metal bar is placed inside a protective case during transportation. The masses of the protective cases are modelled as 30% of the masses of the metal bars.

Let T denote the total mass of 4 of the metal bars and their protective cases. Find the variance of T. [2]

[RVHS 2025 Prelim P2 Q12]

Solution		Comments
(a) **[1]**		

(b) **[2]**	Let D be the random variable denoting the mass of a randomly chosen metal disc. $D \sim N(18,\ 0.6^2)$ Probability required $= \binom{2}{1} [P(D<19)]\,[P(D>19)] - - - (*)$ $= 0.0910128294$ ≈ 0.0910 (3 s.f.)	
(c) **[2]**	$P(18-k<D<18+k)<0.9$ Since $P(17.01308782<D<18.98691218)=0.9$, then $18+k<18.98691218$ $k<0.987$	
(d) **[2]**	7.65% of the metal bars have masses more than 21 grams. Let B be the random variable denoting the mass of a randomly chosen metal bar. $B \sim N(20, \sigma^2)$ $P(B>21)=0.0765$ $P\left(Z>\frac{21-20}{\sigma}\right)=0.0765$ $P\left(Z>\frac{1}{\sigma}\right)=0.0765$ $\frac{1}{\sigma}=1.429014729$ $\sigma=0.6997$ $\sigma=0.700$ (3.s.f)	

(e) **[3]**	Let S be the random variable denoting the mass of a randomly chosen metal strip. $S = 0.8B \sim N(0.8 \times 20,\ 0.8^2 \times 0.5^2)$ $S \sim N(16,\ 0.4^2)$ $P(15.8 < S < 17.2) = 0.690125006$ Let X be the random variable denoting the number of metal strips whose mass lies between 15.8 grams and 17.2 grams, out of 8 bars. $X \sim B(8,\ 0.690125006)$ $P(X \geq 5)$ $= 1 - P(X \leq 4)$ $= 0.7875986163$ ≈ 0.788 (3.s.f)  8 strips, each "success" if $15.8 < S < 17.2$ (prob. 0.690125006): $X \sim B(8,\ 0.690125006)$, need $X \geq 5$	

(f) **[2]**	Let C be the random variable denoting the mass of a randomly chosen protective case. $C \sim N(0.3\times 20,\ 0.3^2\times 0.5^2)$ $C \sim N(6,\ 0.15^2)$ $B \sim N(20,\ 0.5^2)$ Let $Y = B + C$ $Y \sim N(20+6,\ 0.5^2+0.15^2)$ $Y \sim N(26,\ 0.2725)$ Let $T = Y_1 + Y_2 + Y_3 + Y_4$ $\text{Var}(T) = 4\times 0.2725$ $= 1.09$ (3 s.f) **Common Mistake** Let C be the random variable denoting the mass of a randomly chosen protective case. $C = 0.3B$ $B \sim N(20,\ 0.5^2)$ Let $Y = B + C$ $Y = B + 0.3B = 1.3B$ $Y \sim N(1.3\times 20,\ 1.3^2\times 0.5^2)$ $Y \sim N(26,\ 0.4225)$ Let $T = Y_1 + Y_2 + Y_3 + Y_4$ $T \sim N(4\times 26,\ 4\times 0.4225)$ $T \sim N(104,\ 1.69)$	